Timeline of System Design Interview

System Design Interview :- $\boxed{45 \text{min}}$

1. First 5 minutes :- Introduction
2. Next 5 minutes :- General Problem statement to design something
{ Your target should be to get clarity on the question that is asked }

3. System Design Interview will be open ended and you are going to be the driver of that interview.

4. While you try to get the clarity, you need to design the system according to the requirement which your interviewer states you.

5. Functional Requirement :-

(i) Think of the feature that product has got { Instagram ⇒ Upload Photos, Follow others, Messaging, Reels, Timeline Generation }

6. Non Functional Requirement (NFR):-

- Technique that you need to follow here is: don't be $\boxed{\text{SCARED}}$

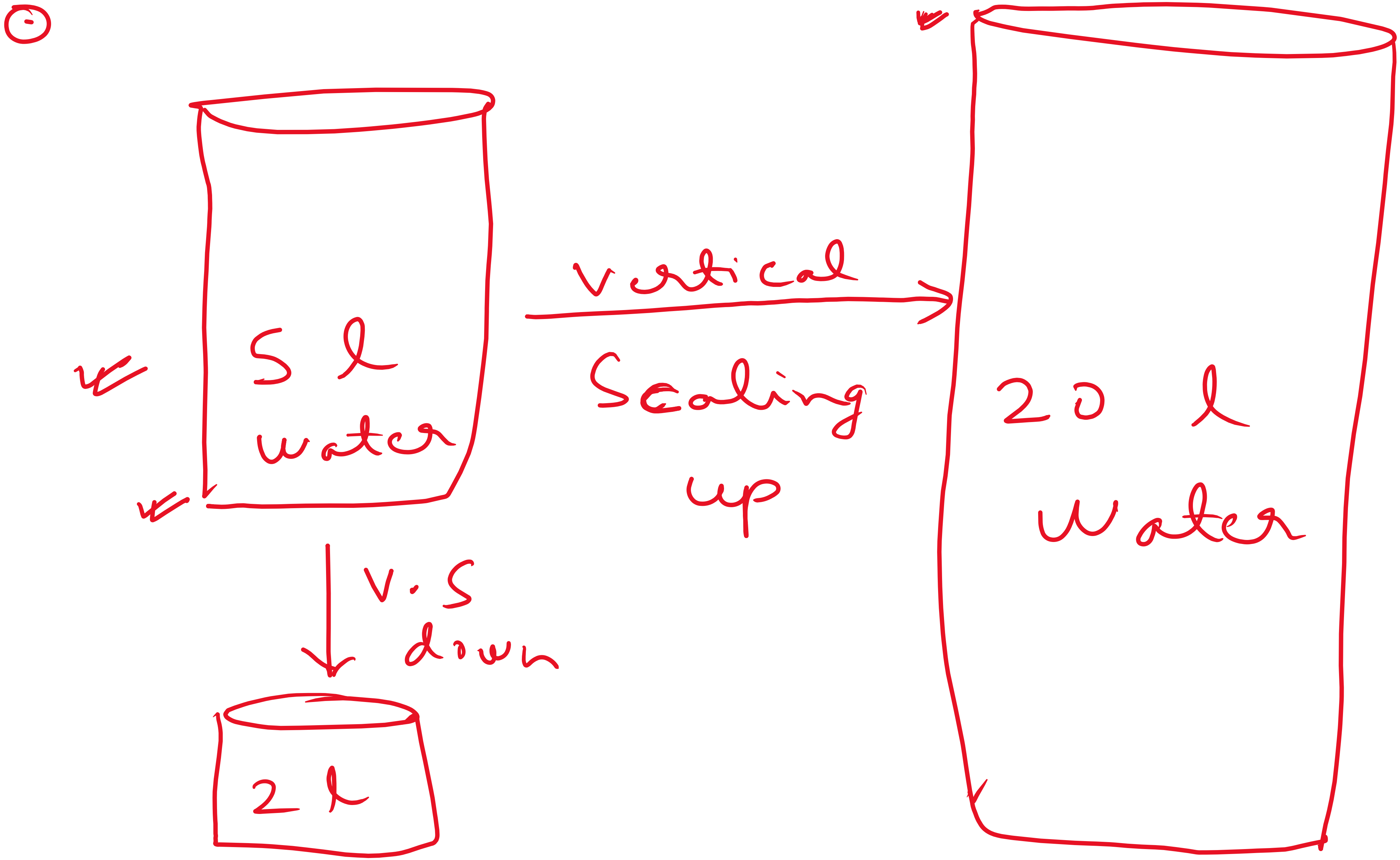
- SCARED

1. <u>Scalability :-</u>

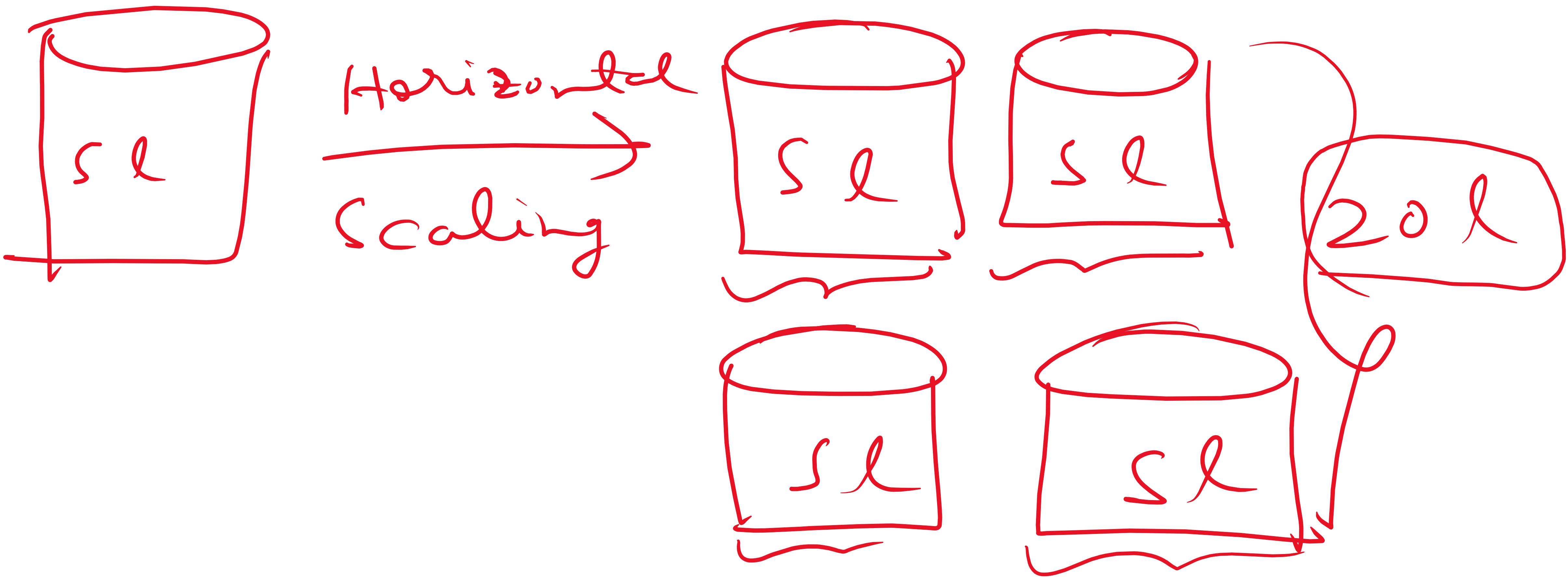
(i) S → System should be scalable.

(i) Horizontal Scaling

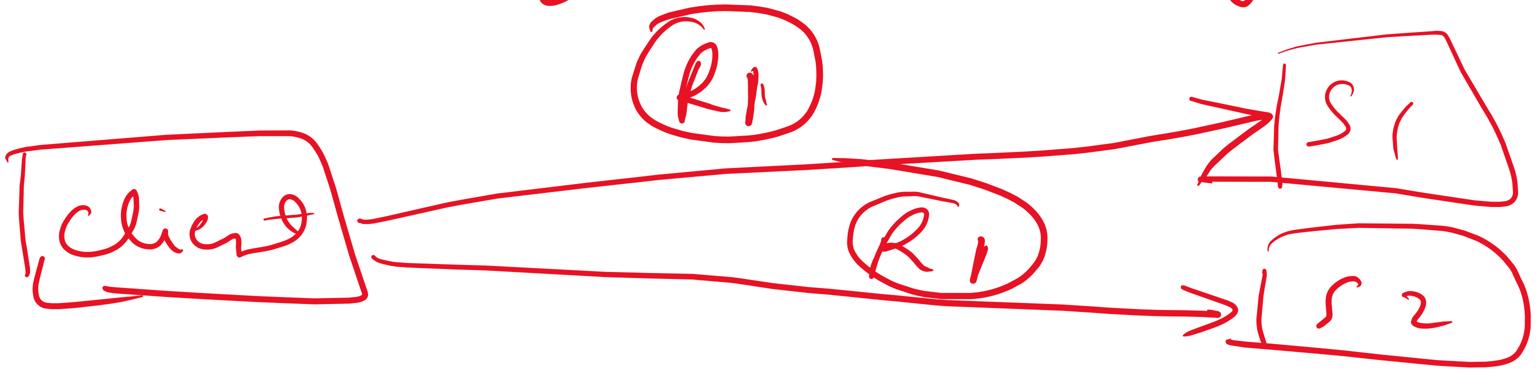
(ii) Vertical Scaling.


5 l
water
vertical
Scaling
up
20 l
water
V.S
down
2 l

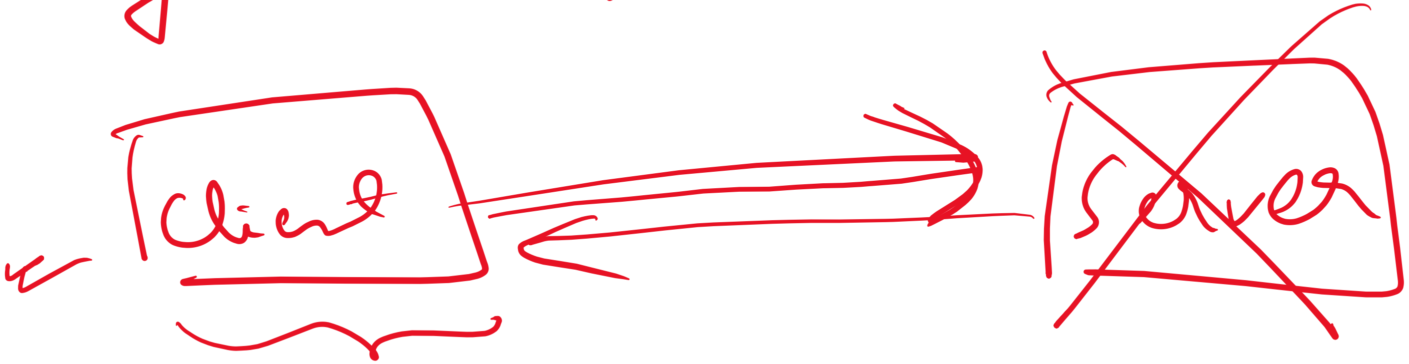
(iii) Horizontal Scaling :-



① Consistency :- All the system or instances that you are going to use in your design should show the same value or response when requested by client.



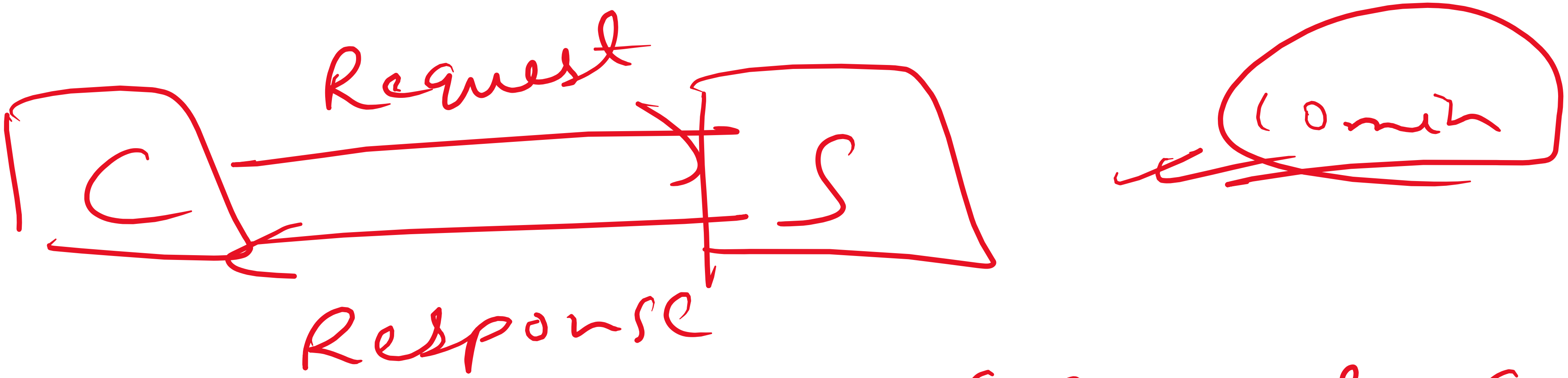
○ Availability :- the resource is accessible or available for the client or users whenever they are making request without any downtime.

○ 

○ R → Reliability :- the system or resource performs consistently and accurately delivering the results as expected by clients with minimal risk of failure.

- **Latency** :- It refers to the delay or the time it takes for the data to travel from one machine (resource or server) to the other machine (client).

- Low latency is also an important NFR. (Non Functional Requirements)



(Capacity Estimation)

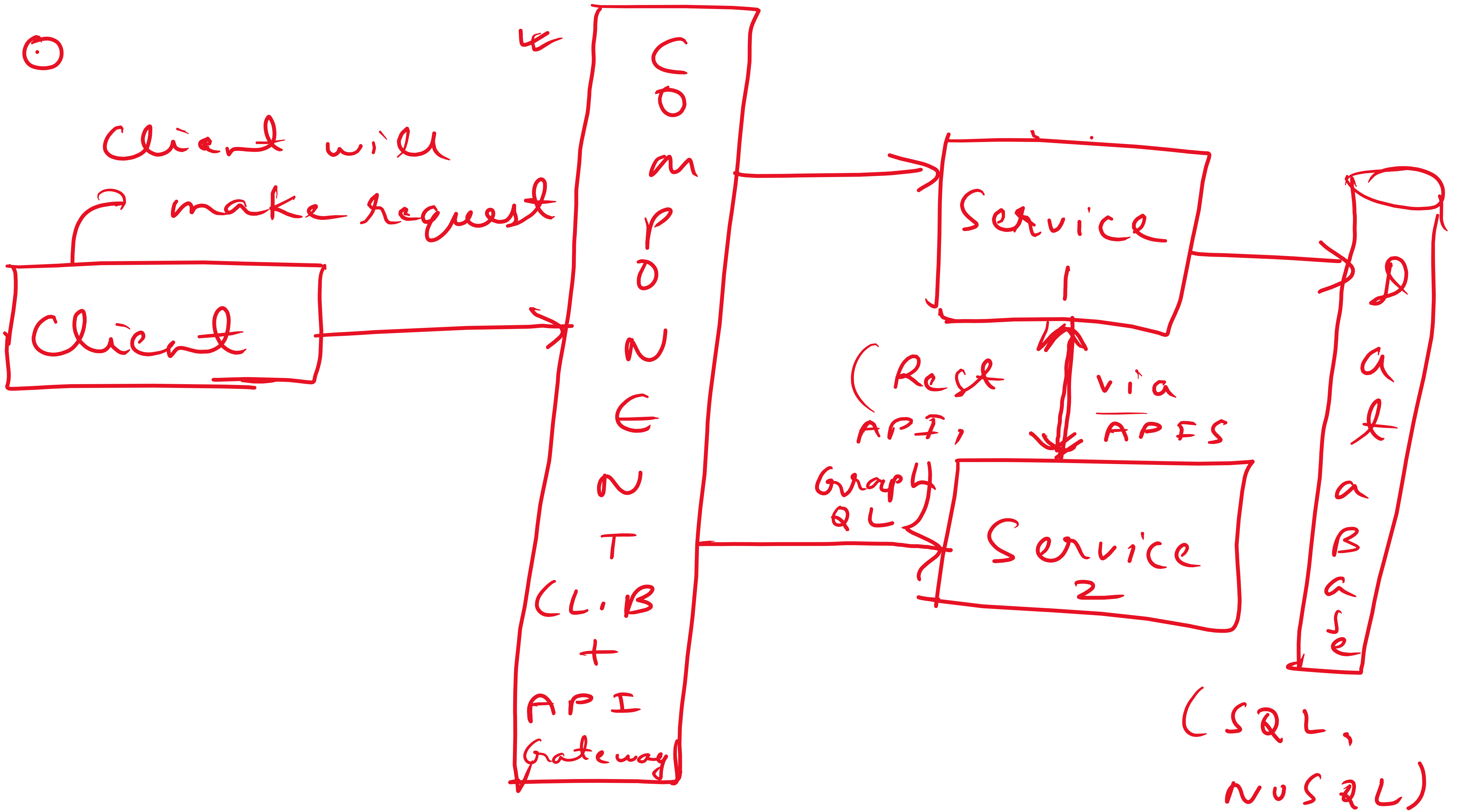
① Within next 2 minutes :-

1. Ask your Interviewer related Daily Active User (D.A.U)

2. Monthly Active User (M.A.U)

- DAU / MAU will help you in figuring out the scalability of your system.

- After you have got the clarification, now you need to start building your system which will meet your functional Requirement (FR), NFR and Capacity Estimation.


Client will make request
Client
COMPONENT
(CLB + API Gateway)
Service 1
Service 2
via APIS
(Rest API, GraphQL)
Database
(SQL, NoSQL)